Note that the exact error equals

$$M - Q(h) = e - 2.7525\ldots = -0.0342\ldots.$$

In this example the error estimate is very reliable.

To receive a better approximation the error estimate can be added to the approximation:

$$Q(h) + c_p h^p = 2.7525\ldots - 0.0348\ldots = 2.7177\ldots.$$

In the above example, the value of p was computed using Richardson's extrapolation. However, using Theorem 3.2.1, it is clear that $p = 1$, and this value could have been used immediately in equation (3.13b) in order to determine $c_p h^p$. In practice, more complex situations are found, and the following complications may occur:

- It is not known whether higher-order derivatives exist and/or are bounded.
- The final result is a combination of various approximation methods. The influence of these approximations on p is not always clear.
- During implementation of the algorithm in a computer program, errors may be made.

To reveal any of these complications it is good practice to verify whether the calculated p is close to the p that follows from theory.

3.7.3 Formulae of higher accuracy from Richardson's extrapolation *

In several applications the value of p in (3.10) is known. In that case Richardson's extrapolation can be used to determine formulae of higher accuracy.

This is done by making use of the fact that the error estimates for $Q(h)$ and $Q(2h)$ equal

$$M - Q(h) \;= c_p h^p + \mathcal{O}(h^{p+1}), \tag{3.15a}$$

$$M - Q(2h) = c_p (2h)^p + \mathcal{O}(h^{p+1})\,. \tag{3.15b}$$

Multiplying equation (3.15a) by 2^p and subtracting equation (3.15b) from this yields

$$2^p(M - Q(h)) - (M - Q(2h)) = 2^p(c_p h^p) - c_p(2h)^p + \mathcal{O}(h^{p+1}),$$

such that

$$(2^p - 1)M - 2^p Q(h) + Q(2h) = \mathcal{O}(h^{p+1}).$$

This means that

$$M = \frac{2^p Q(h) - Q(2h)}{2^p - 1} + \mathcal{O}(h^{p+1}). \tag{3.16}$$

The value $(2^p Q(h) - Q(2h))/(2^p - 1)$ is a new approximation formula for M with an accuracy that is one order higher than the order of $Q(h)$.

Example 3.7.2 (Forward difference of higher accuracy)

As an example, the forward-difference method is considered. The error in the forward-difference formula may be written as

$$f'(x) - Q_f(h) = c_1 h + \mathcal{O}(h^2), \tag{3.17}$$

and the difference for $2h$ equals

$$f'(x) - Q_f(2h) = c_1 2h + \mathcal{O}(h^2). \tag{3.18}$$